Mid Term Examination (March-2025)
B.Tech.(CSE/IT/CST/ITE)

Subject: Programming in Python | **Subject Code: CIE-332T**
Time: 1.30 Hrs | **Max. Marks = 30**

Note: Attempt three questions including Ques 1 which is compulsory

Q1. What would be output of the following code, Explain your answer: **2x5=10 Marks**

Choice 1

a)
```
import sys
def check(x):
    if x % 2 == 0:
        sys.exit()
    return x * 2
print(check(5) + check(3))
```
(CO1)

b)
```
x = 10
y = 20
z = (x < y) and (y > 15) or (x == 10) and not (y == 20)
print(z)
```
(CO1)

c)
```
def func(a, b=5):
    return a * b
x = func(3)
y = func(2, 10)
print(x + y)
```
(CO1)

d)
```
def compute(x, y=5, z=10):
    return x * y + z
result1 = compute(2)
result2 = compute(3, 4)
result3 = compute(1, z=20)
print(result1, result2, result3)
```
(CO2)

e)
```
word_list = ["hello", "world", "python"]
def modify_word():
    global word_list
    word_list[1] = word_list[1].upper()
    local_list = word_list[:2]
    return "-".join(local_list)
print(modify_word())
print(word_list)
```
(CO2)

OR

Choice 2

a)
```
x = "Automate"
y = "Python"
z = (x[:4] + y[-3:]) * 2
print(z[::3] + z[-2] + z[5])
```
(CO1)

b)
```
def test(n):
    if n % 2 == 0:
        return "Even"
    elif n % 3 == 0:
        return "Divisible by 3"
    return "Odd"
print(test(9) + " " + test(12))
```
(CO1)

c)
```
import pprint
data = {'name': 'Alice', 'age': 25, 'city': 'New York'}
pprint.pprint(data)
```
(CO2)

d)
```
d = {1: 'one', 2: 'two', 3: 'three'}
keys_list = list(d.keys())[::-1]
values_list = list(d.values())[::-1]
print(keys_list[0], values_list[1])
```
(CO2)

e)
```
data = {'Alice': 25, 'Bob': 30, 'Charlie': 35}
try:
    age = data['David']
except KeyError:
    age = sum(data.values()) // len(data)
print(age)
```
(CO2)

Q2. a. Explain the difference between local and global scope in Python. How does the ***global*** statement affect variable scope? Provide an example where modifying a global variable inside a function leads to unexpected behavior. **3 Marks (CO2)**

(Please write your Exam Roll No.) *Exam Roll No.....*

END TERM EXAMINATION

SIXTH SEMESTER [B.TECH] JUNE 2024

Paper Code:CIE-332T/IOT-330T ***Subject- Programming in Python***

Time: 3hours ***Max. Marks: 75***

Note : Attempt five questions in all including Q. No. 1 which is compulsory. Select one Question from each Unit. Assume suitable missing data if any.

Q1 Answer following in brief. **[5x5 = 25]**

(a) What is the role of data frame? How can data frames be merged?
(b) What are the features of Python Programming language?
(c) Differentiate between list, tuple and dictionary with example.
(d) Write a short note on different methods to read data from a file.
(e) How is exception handling implemented in python?

UNIT-I

Q2 (a) What are loop interruption statements? What is the difference between break, continue and pass? 17. Write a program that prints all integers lying between 1 and 50 that aren't divisible by either 2 or 3. **[6]**

(b) Explain the different numeric, assignment, augmented operators used in python. **[3]**

(c) Explain how operator precedence is used in evaluating expressions. **[3.5]**

Q3 (a) What are the commonly used operators in python? Explain operator overloading with the help of an example. **[6]**

(b) What is type conversion? Write a python program to demonstrate number type conversions. **[6.5]**

UNIT-II

Q4 (a) What are the different ways of string manipulation in python? WAP in python check whether the string is Symmetrical or Palindrome **[6]**

(b) What are functions in python? WAP in python to use Pythagoras theorem to calculate the length of the hypotenuse of a right-angled triangle where the other sides have lengths 3 and 4. **[6.5]**

Q5 (a) Define the following: find(), isalnum(), zfill(), strip(), split() and rindex(). **[6]**

(b) Write the definition of a function Reverse(X). WAP to display the elements in reverse order such that each displayed element is twice of the original element (element *2) of the List X. **[3]**

(c) Write a function mul ( ) which accepts list L, odd_L, even_L as its parameters. Transfer all even values of list L to even_L and all odd values of L to odd_L. **[3.5]**

UNIT-III

Q6 (a) Write a program that accepts filename as an input from the user. Open the file and count the number of times a character appears in the file. **[6]**

(b) What are files? Why do we need them? What are different access modes in which you can open file? **[6.5]**

Q7 (a) What is directory access? How do you walk through a directory tree? **[6]**

(b) Write a python program to merge two files and then change the content of this file to upper case letters. **[6.5]**

P.T.O.

MID TERM EXAMINATION

SIX SEMESTER [B. TECH.] JAN 2024

Paper Code: CIE-332T **Subject: Programming in Python**

Time: 1Hour 30 Min. **Maximum Marks: 30**

Note: Attempt three questions in all including Q. No. 1 which is compulsory.

Q1. (a)	What is the difference between logic error and syntax error?	[2]	CO1
(b)	What are the features of Python Programming language?	[2]	CO1
(c)	Write a Python program to replace last value of tuples in a list. Sample list: [(10, 20, 40), (40, 50, 60), (70, 80, 90)] Expected Output: [(10, 20, 100), (40, 50, 100), (70, 80, 100)]	[2]	CO3
(d)	Create a 5x 2 integer array from a range between 100 to 200 such that the difference between each element is 10.	[2]	CO2
(e)	What is lamda function? Explain with example.	[2]	CO2
Q2. (a)	Write a program to calculate GCD of two numbers.	[5]	CO2
(b)	What are the different types of decision control statements? Explain with examples.	[5]	CO1
Q3. (a)	Write a Python program to calculate the number of minutes in a week. Make variables for DaysPerWeek, HoursPerDay, and MinutesPerHour.	[5]	CO2
(b)	What will be the output of the following Python code snippet? for i in ".join(reversed(list('abcd'))): print (i)	[2]	CO2
(c)	What is the difference between List, Tuples and Dictionary? Explain in detail.	[3]	CO3
Q4. (a)	Write a program to demonstrate the concept of exception handling in python.	[5]	CO4
(b)	Justify the statement: "Strings are immutable". Explain string manipulation in python programming with the help of code.	[5]	CO3

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